Development of an LCD Spatial Light Modulator and Exploration of Transport-of-Intensity Imaging

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Abstract

This project investigates the development of a low-cost liquid crystal display (LCD)–based spatial light modulator (SLM) for optical applications. A consumer LCD was modified into a transmissive SLM and integrated into a laser-based system to generate programmable light patterns using microcontroller control. The Transport-of-Intensity Equation (TIE) was applied to reconstruct the phase distribution of the LCD surface from defocused intensity images processed in MATLAB. This approach enables phase reconstruction and surface characterization using low-cost optical hardware.

Introduction

Liquid crystal displays (LCDs) can function as spatial light modulators (SLMs) by controlling light at the pixel level [1]. While commercial SLMs provide precise phase control and high optical quality, their high-cost limits accessibility for smaller laboratories and educational settings [2]. Consumer LCDs offer a low-cost alternative, but their phase behavior and surface uniformity are not well characterized for precision optical applications [1]. This project investigates the use of a modified LCD as a spatial light modulator and applies the Transport-of-Intensity Equation (TIE) to reconstruct the optical phase and analyze surface irregularities.

Methodology

Experimental Setup

The LCD module was modified by removing the diffuser, backlight, and metal backing layers to expose the liquid crystal panel for transmissive spatial light modulation. The modified LCD was integrated into a laser-based optical system consisting of a laser source, three lenses (including an imaging lens), and a camera aligned along a linear optical rail. The camera lens was removed to allow direct imaging of the transmitted laser patterns produced by the LCD. All components were mounted on a vibration-damped optical table to maintain alignment and ensure stable image acquisition.

Methodology

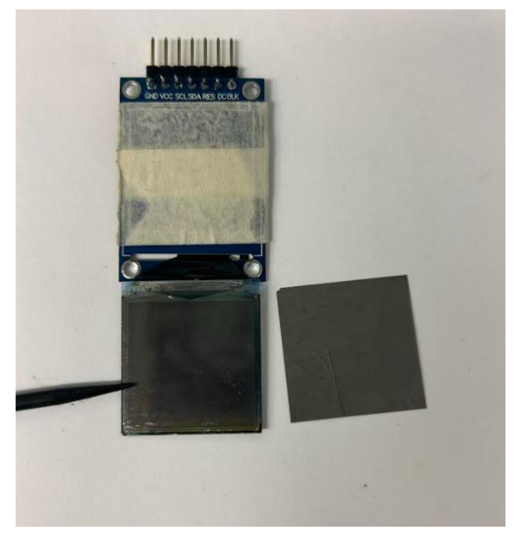

Figure 1:LCD display after removal of diffuser and backlight layers, exposing the liquid crystal layer used for transmissive spatial light modulation.

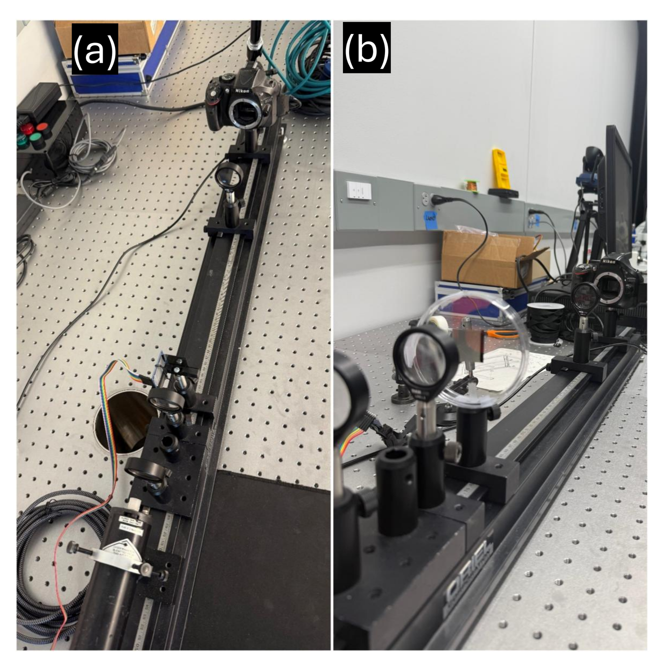

Figure 2: Optical setup used for the experiment. (a) Optical rail with three lenses, a camera, and the LCD connected for spatial light modulation testing. (b) LCD sample positioned in the optical path for Transport-of-Intensity Equation (TIE) measurements.

TIE Phase Reconstruction

The Transport-of-Intensity Equation (TIE) relates axial intensity variations to the phase of an optical wavefront:

$$\frac{\partial I(x,y,z)}{\partial z} = -\frac{\lambda}{2\pi}\nabla_{\perp}\cdot[I(x,y,z)\nabla_{\perp}\phi(x,y,z)]$$

Defocused intensity images were captured by moving the imaging lens slightly before and after the focal plane (z^{-}, z^{+}) .The axial intensity derivative was approximated using

$$\frac{\partial I}{\partial z} \approx \frac{I_{+} - I_{-}}{2dz}$$

The equation was rewritten in Poisson form and solved in MATLAB using a Fourier-domain approach (FFT) to recover the phase distribution. The reconstructed phase was then converted into surface height using

$$h(x,y) = \frac{\lambda}{4\pi}\phi(x,y)$$

producing a quantitative 3D surface ma**p** from intensity-only measurements. [3]

Results

LCD Spatial Modulation

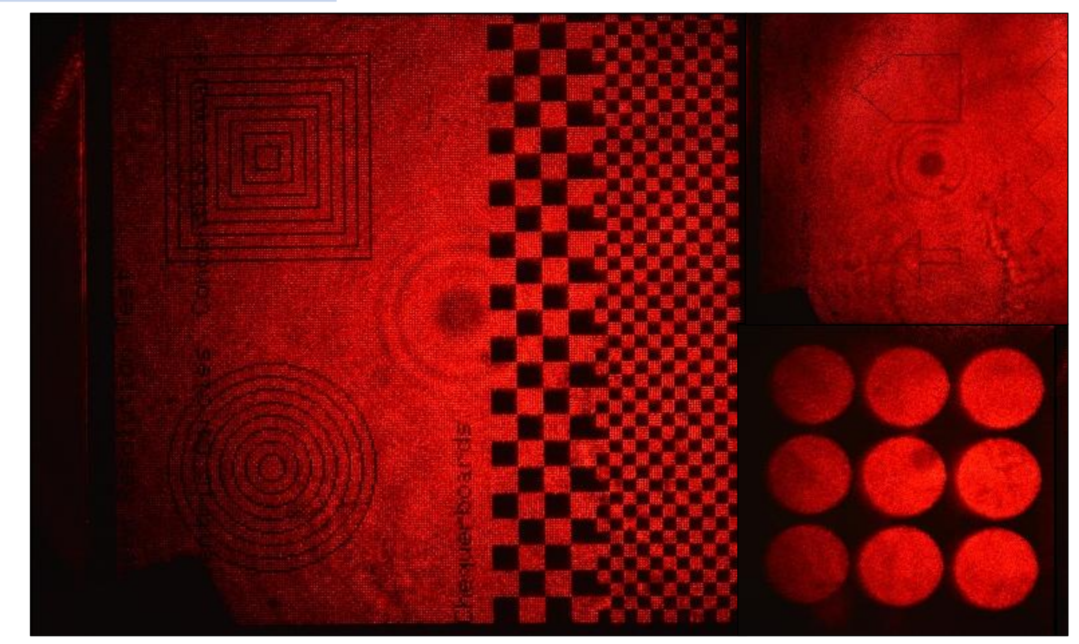

Figure 3: Spatial light modulation produced by Arduino-controlled patterns displayed on the LCD. Under laser illumination, the programmed patterns become visible through the display.

TIE Recontruciton (Normal LCD SLM)

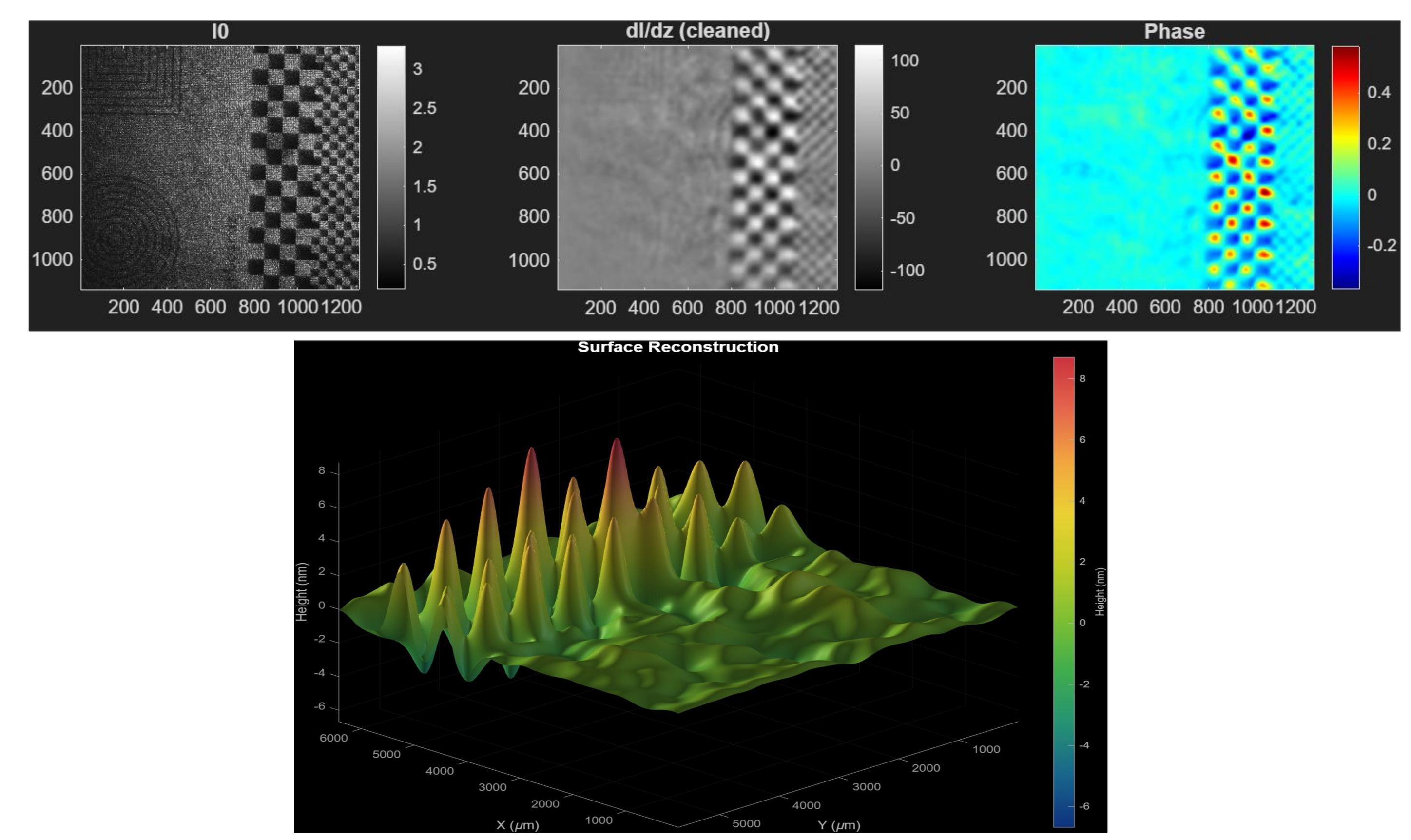

Figure 4: TIE phase reconstruction of a normal LCD region. Intensity images are used to compute the phase distribution and reconstruct the 3D surface profile.

TIE Recontruciton (Damaged LCD SLM)

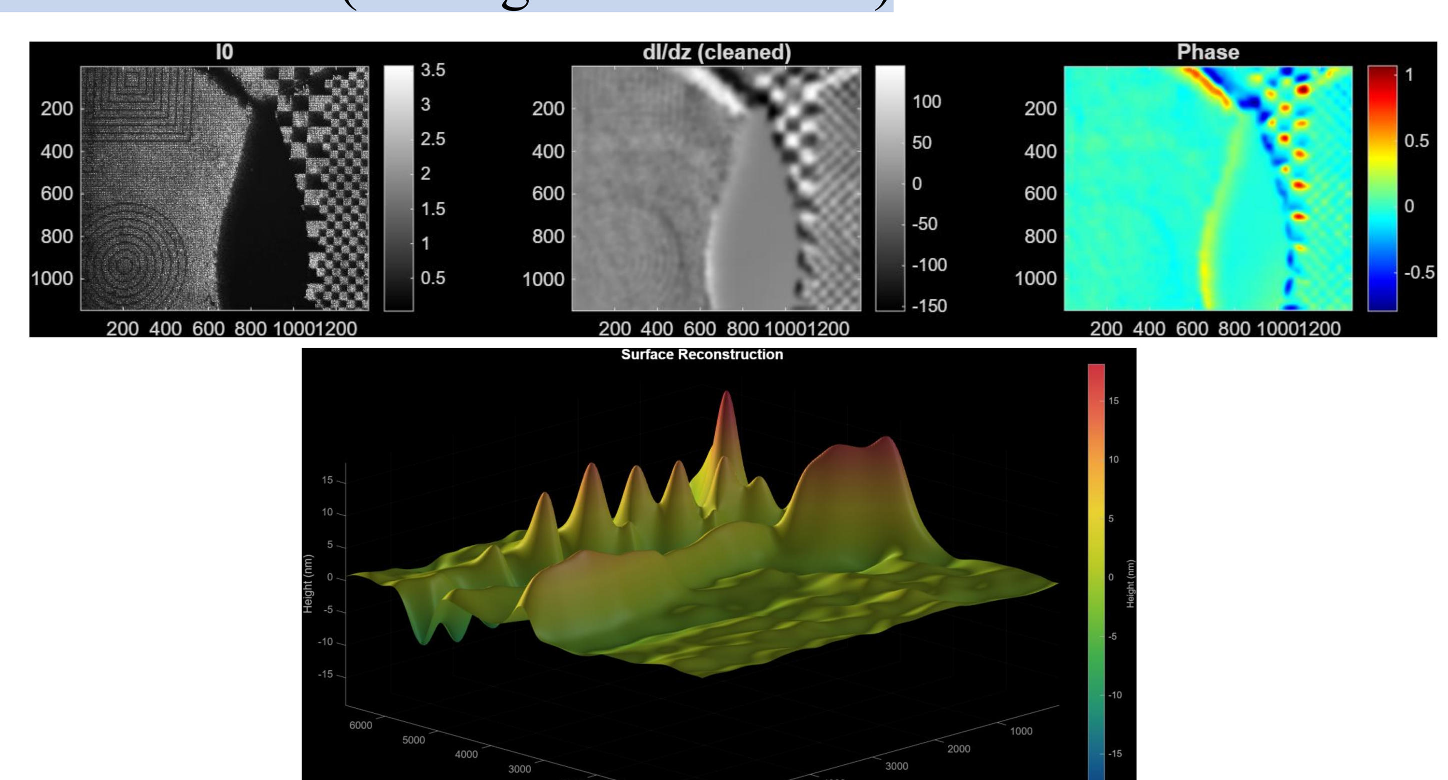

Figure 5: TIE phase reconstruction of a damaged LCD region showing larger phase distortions and increased surface height variations.

SLM turned off Reconstruction (Damaged LCD)

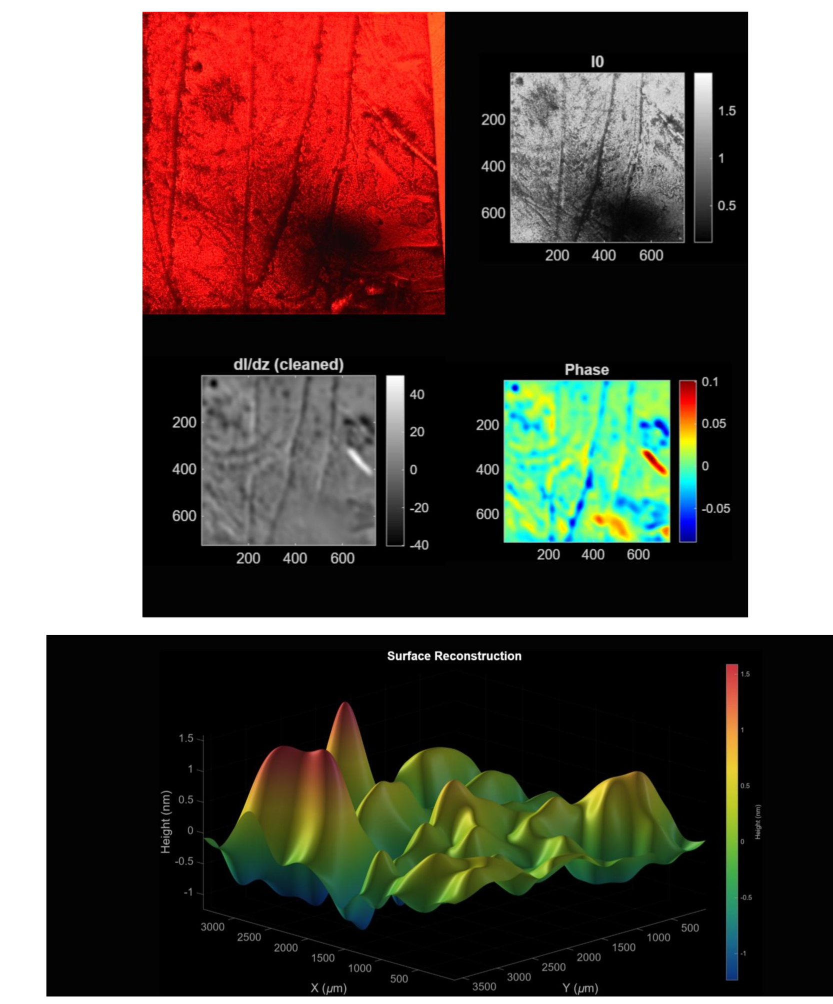

Figure 5: Intensity images of a damaged LCD region with the SLM turned off, showing structural defects used for TIE phase reconstruction

Conclusions/ Future Directions

- A consumer LCD was successfully converted into a transmissive spatial light modulator.
- Arduino control enabled programmable spatial light patterns.
- The Transport-of-Intensity Equation successfully reconstructed phase information from intensity-only measurements.
- Damaged LCD regions produced larger phase distortions and surface height variations.
- Future work will focus on improving modulation resolution and investigating additional LCD panels for phase stability.

References

1. Zhao et al., 2018, *Optics and Lasers in Engineering*.
2. Márquez & Lizana, 2019, *Applied Sciences*.
3. Zuo et al., 2018, *Optics and Lasers in Engineering*.

Acknowledgements

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